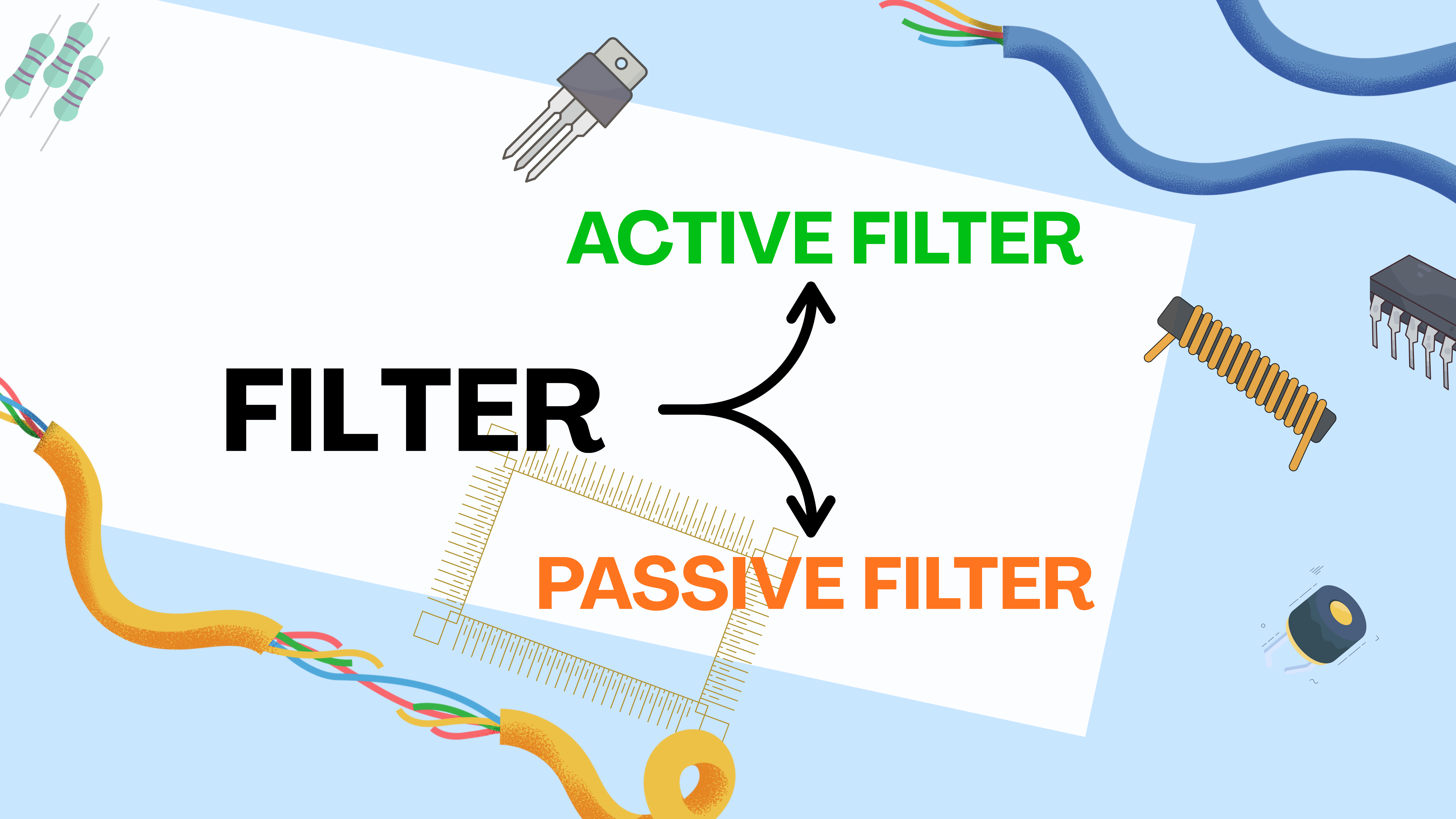
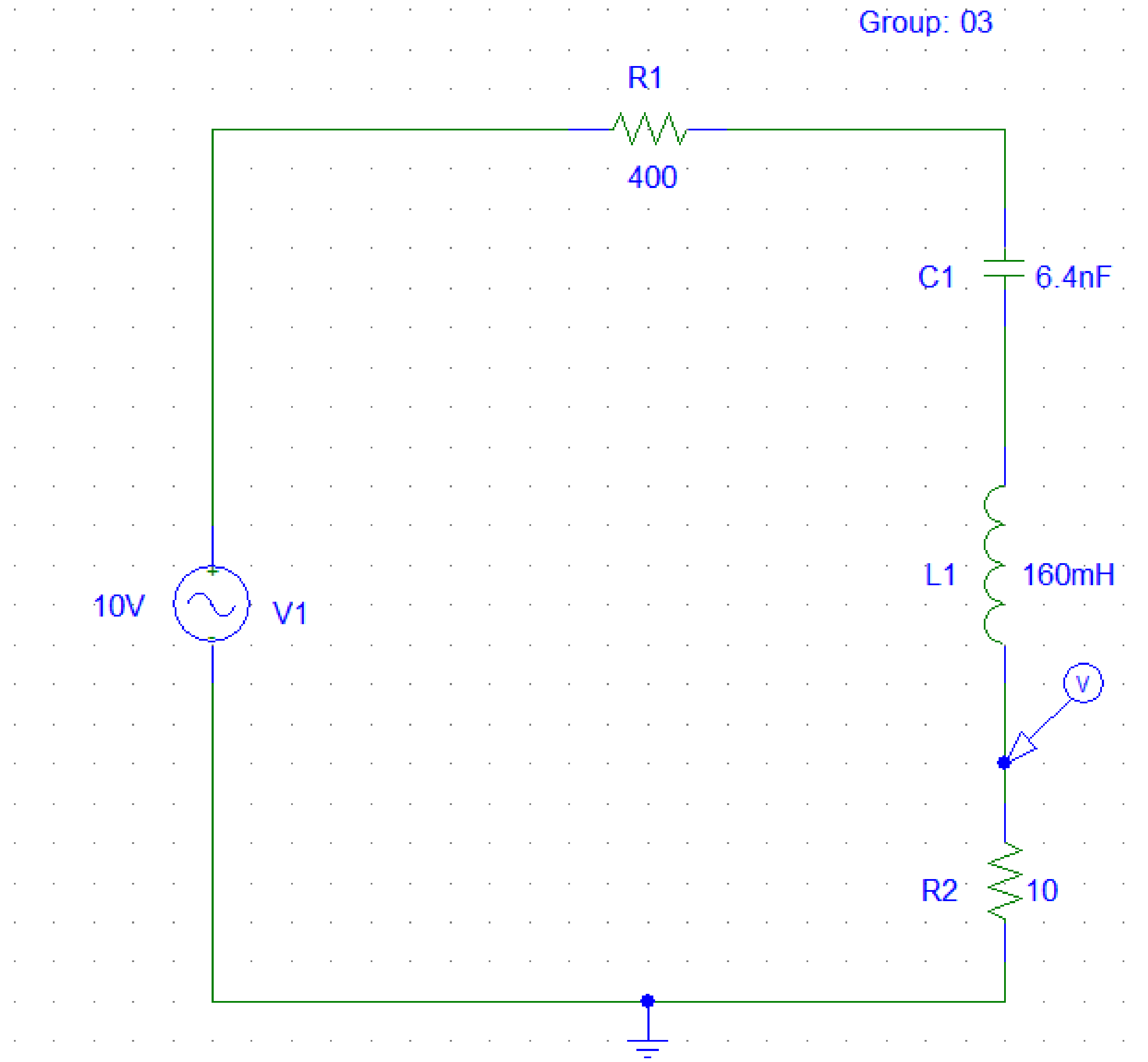
EEE203L

LAB PROJECT


ACTIVE FILTER
FILTER
PASSIVE FILTER



Passive Filter

Resistor (R), Inductor (L), and Capacitor (C).
Do not need an external power supply.

Example: Series RLC circuit .

Active Filter

R, L, C + active components like
Operational Amplifiers
Transistors
Need a DC power supply to operate.
Example: Op-Amp based Low-Pass or High-Pass filters.

Building the circuit

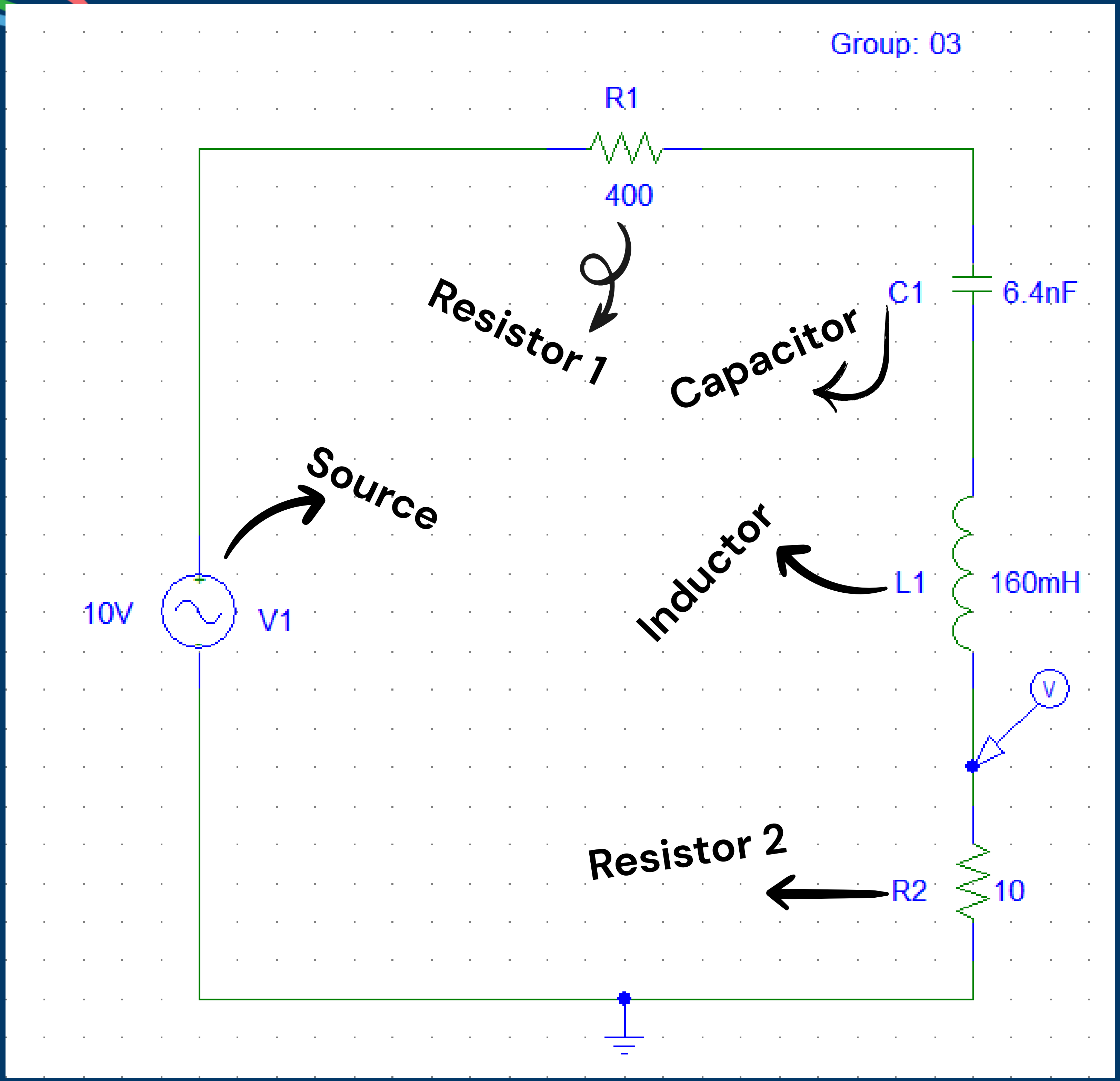
Source: 10 Volt AC voltage source.

Resistors: 2 resistors , R1 at 400 ohms and R2 at 10 ohms.

Capacitors: 6.4 nano Farrad capacitor.

Inductor: 160 milli Henry inductor.

Every element is connected in series, ensuring equal current flow through the circuit.

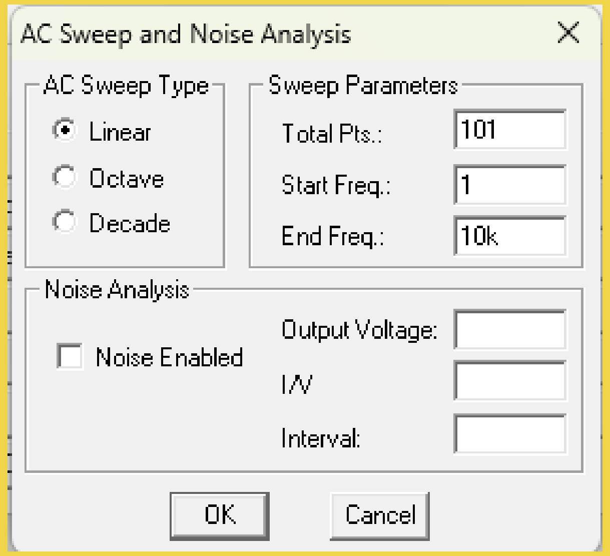


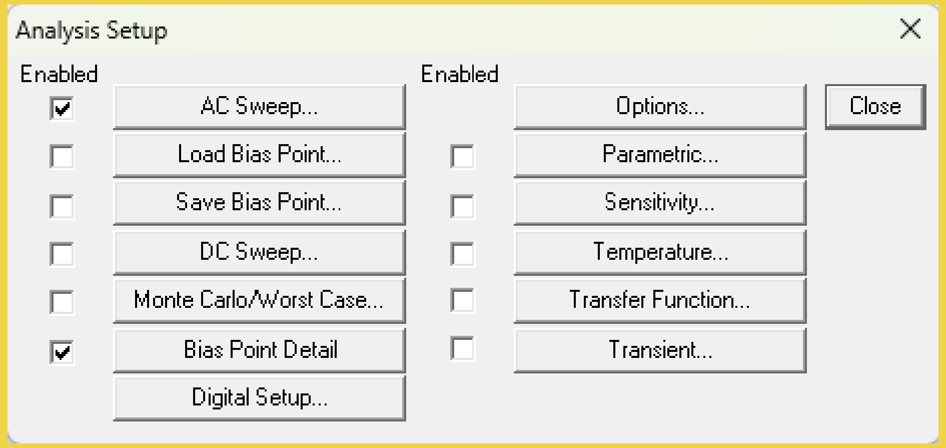
Rechecking the parameters

1.Ensuring the values of the elements:

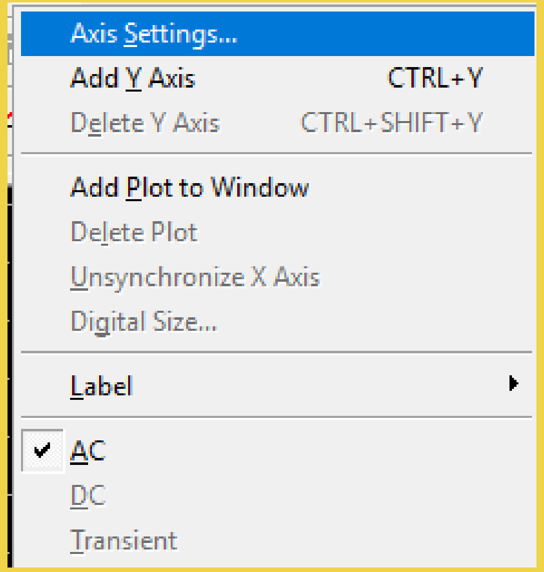
V1 (AC) = 10∠0° V;
R1 = 400 Ω;
R2 = 10 Ω;
C = 6.4 nF;
L= 160 mH;

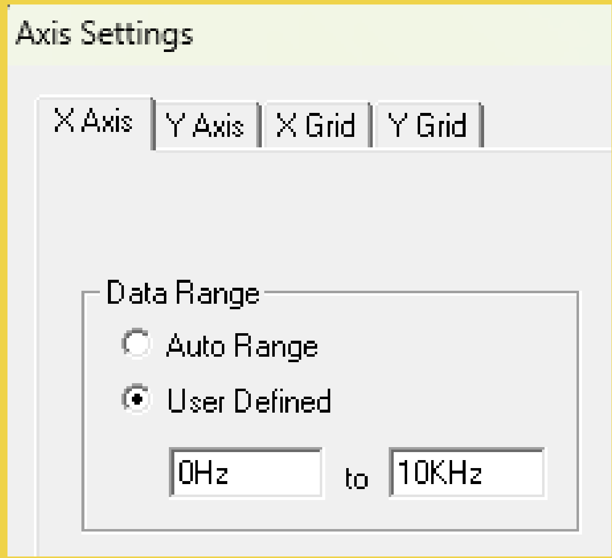
2.setting up the simulation:





3.Setting up the plot:



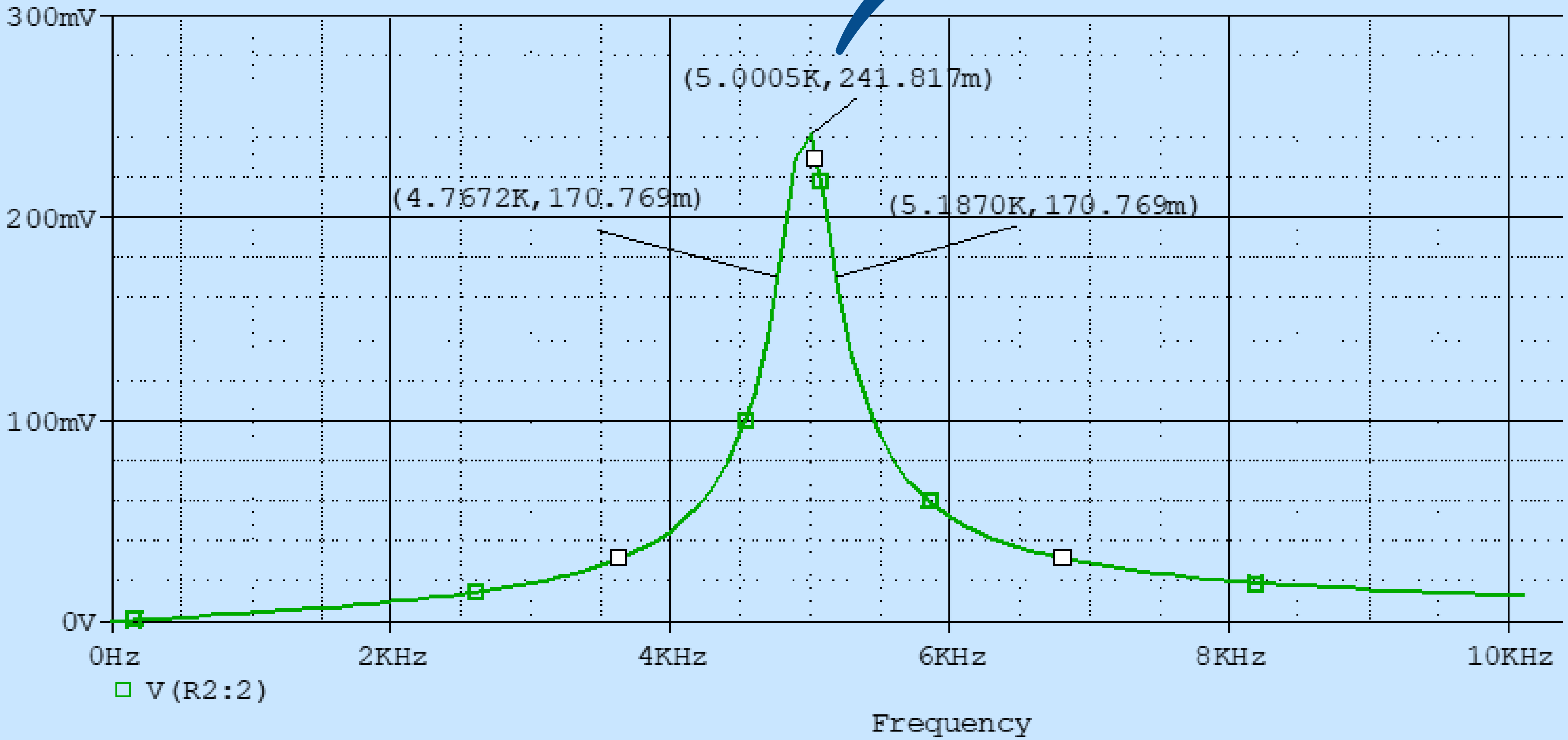


Illustrating the plot

V_0 vs f

A graph that plots V_0 (R_2) at y axis and frequency (f) on the x axis

maximum voltage at
f= 5.0005kHz



Here, the plot shows that, the at the resonant frequency voltage on R_2 which shows that, X_L and X_C will cancel each other out, hence the resistor will remain as the only impedence in the circuit and therefore, the voltage will be at a maximum

e. Complete the following data table:

The maximum value of $\|V_o\|$ (V)	Resonant frequency, f (kHz)	The value of the quantity that identifies the cut-off frequency	The cut-off frequency (kHz)	The Bandwidth (kHz)	Selectivity

Maximum possible output from this circuit from varying frequencies

The frequency at which the output reached its peak

Multiplying peak with 0.707 to estimate where the cut-off may be

Pinpointing the cut-off at both end using the mentioned value

Figuring a range.

From low end to high end.

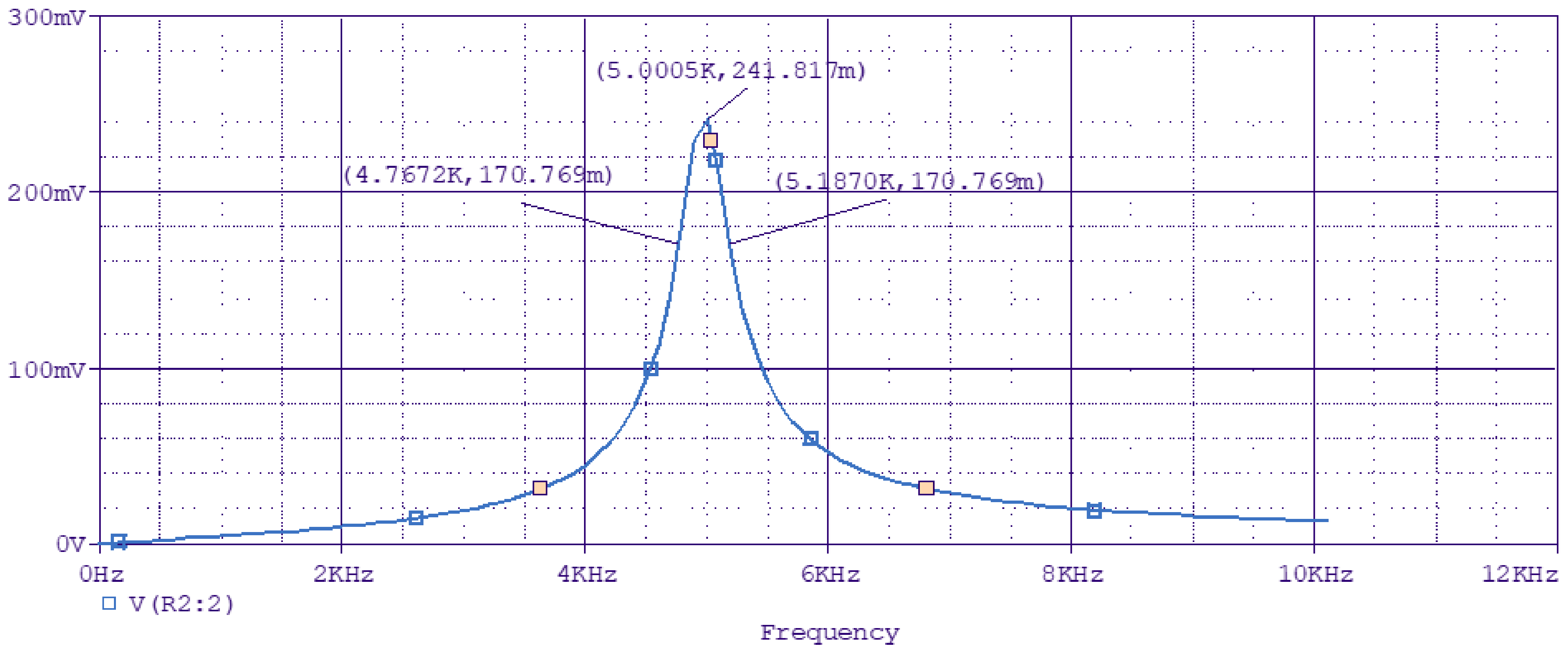
Quality factor

Indicates how sharp the curve is

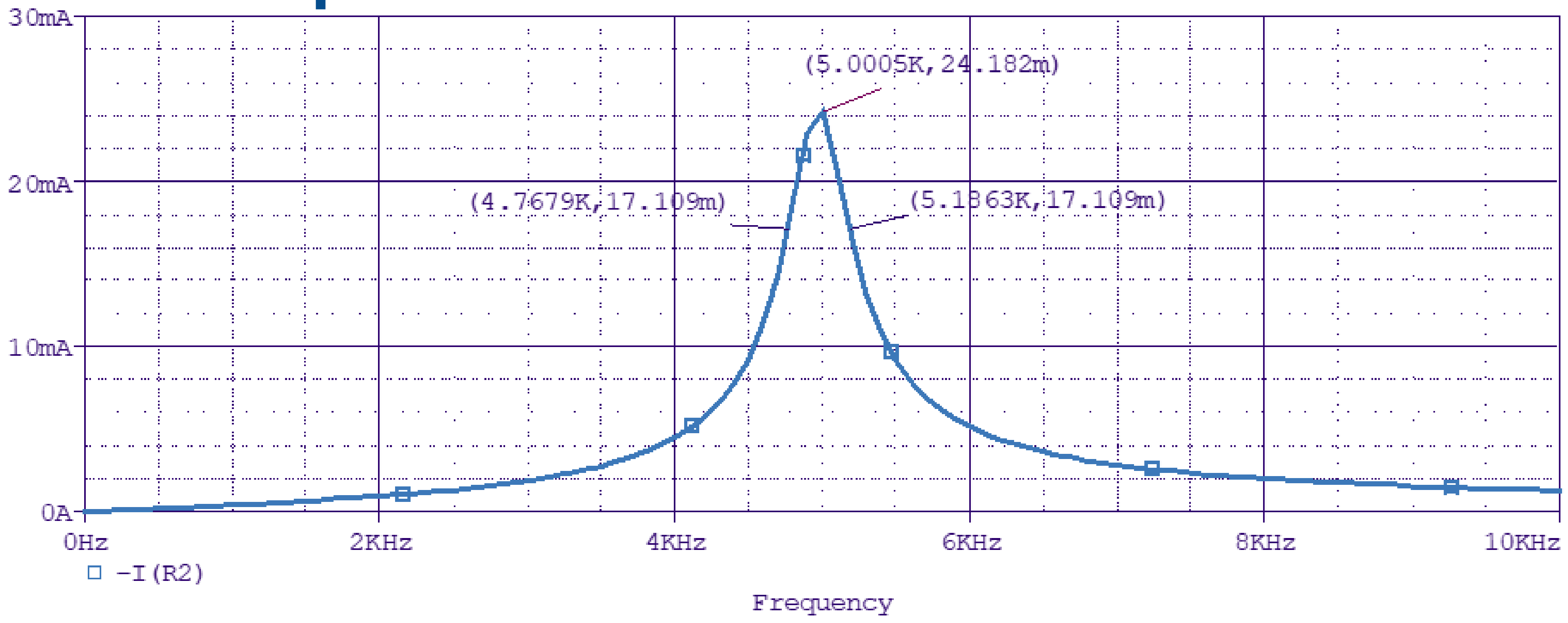
ANSWER:

V_o	0.241817 V
f	5.0005 kHz
0.707 * peak (I_o or V_o)	17.096 mA or 170.769 mV
Cut-off Freq.	f_{cl} = 4.7679 f_{ch} = 5.1863
Bandwidth	0.4184
Selectivity	11.9515

V vs f Graph



-I vs f Graph



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TYPES OF FILTERS

Low-pass

lets low frequencies pass, blocks high.

High-pass

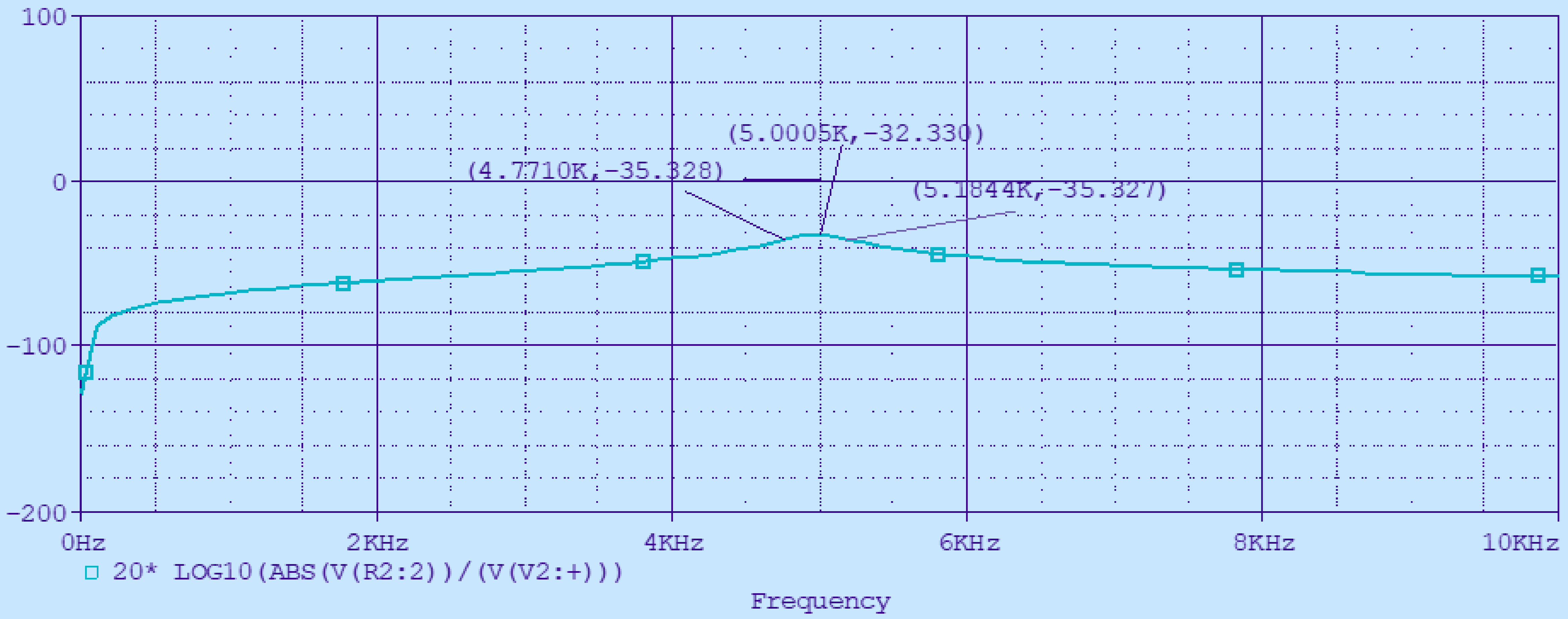
lets high frequencies pass, blocks low.

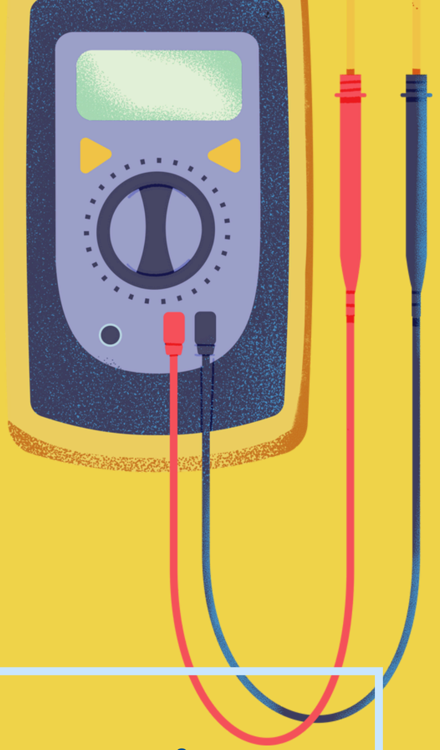
Band-pass

lets only a middle range of frequencies pass.

Band-stop

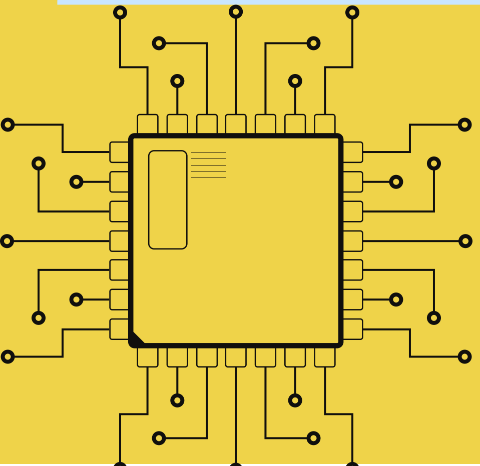
blocks a middle range, lets low & high frequencies pass.



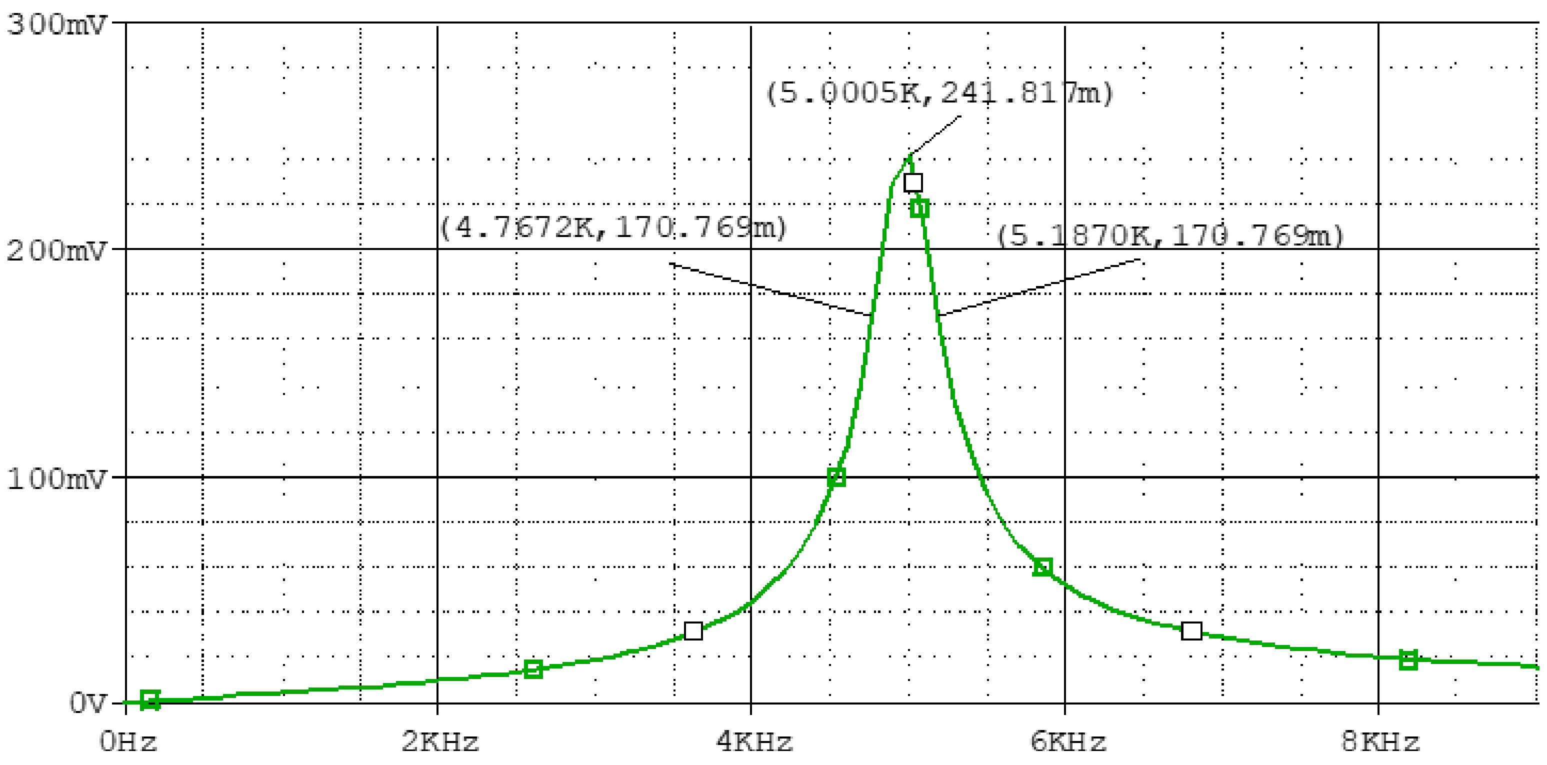


PASS-BAND & STOP-BAND

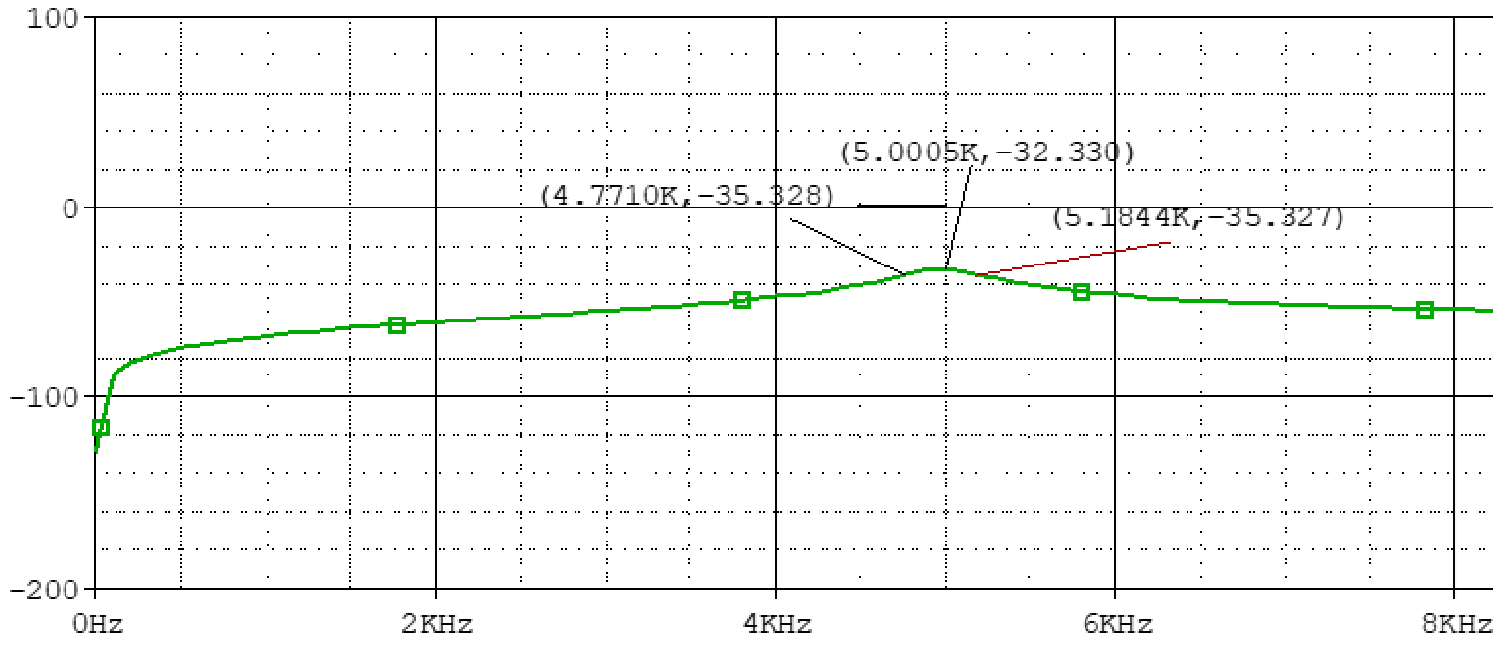
Pass-band	**Range of frequencies where output is above –3 dB.**
Stop-band	**Frequencies where output is below –3 dB.**
For our project	**Pass-band:** **4.7678 kHz – 5.1863 kHz** **Stop-band:** **f < 4.7678 kHz and f > 5.1863 kHz**



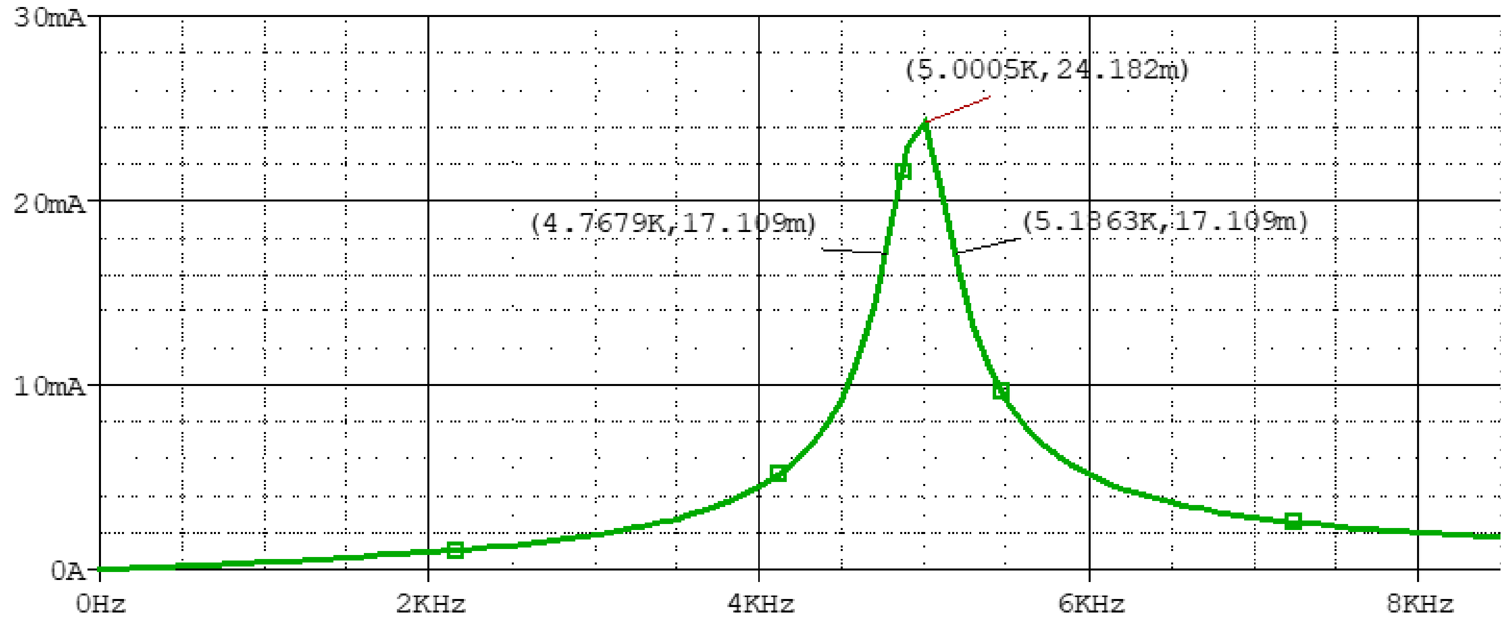
–3 dB, 0.707 Vmax, and 0.707 Imax are equivalent because the signal is reduced to about 70.7% of its maximum amplitude



Graph: $|V_o|$ vs f



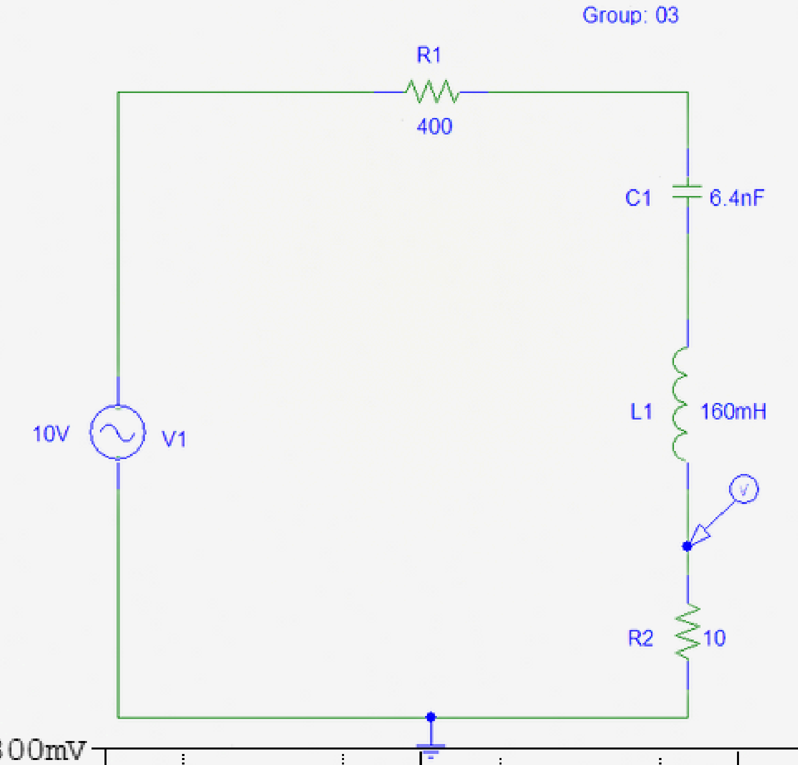
Graph: Gain(db) vs f

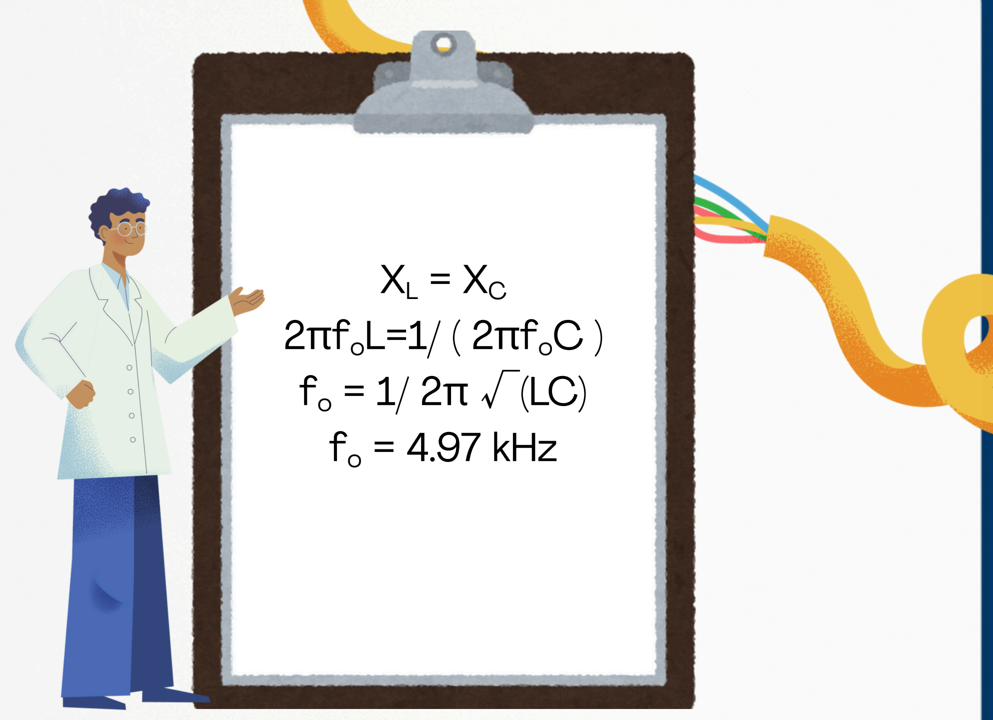


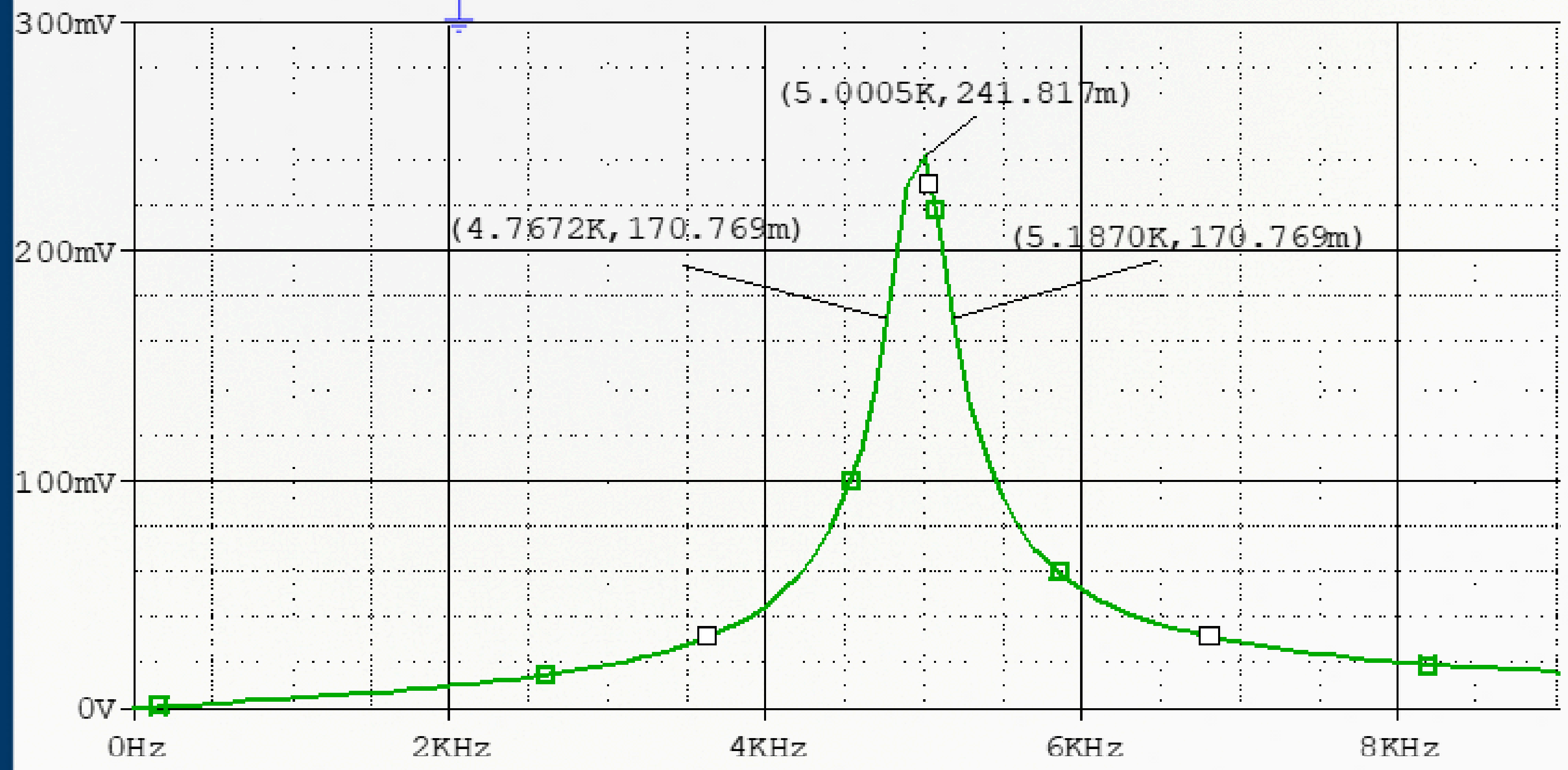
Graph: $|I_o|$ vs f

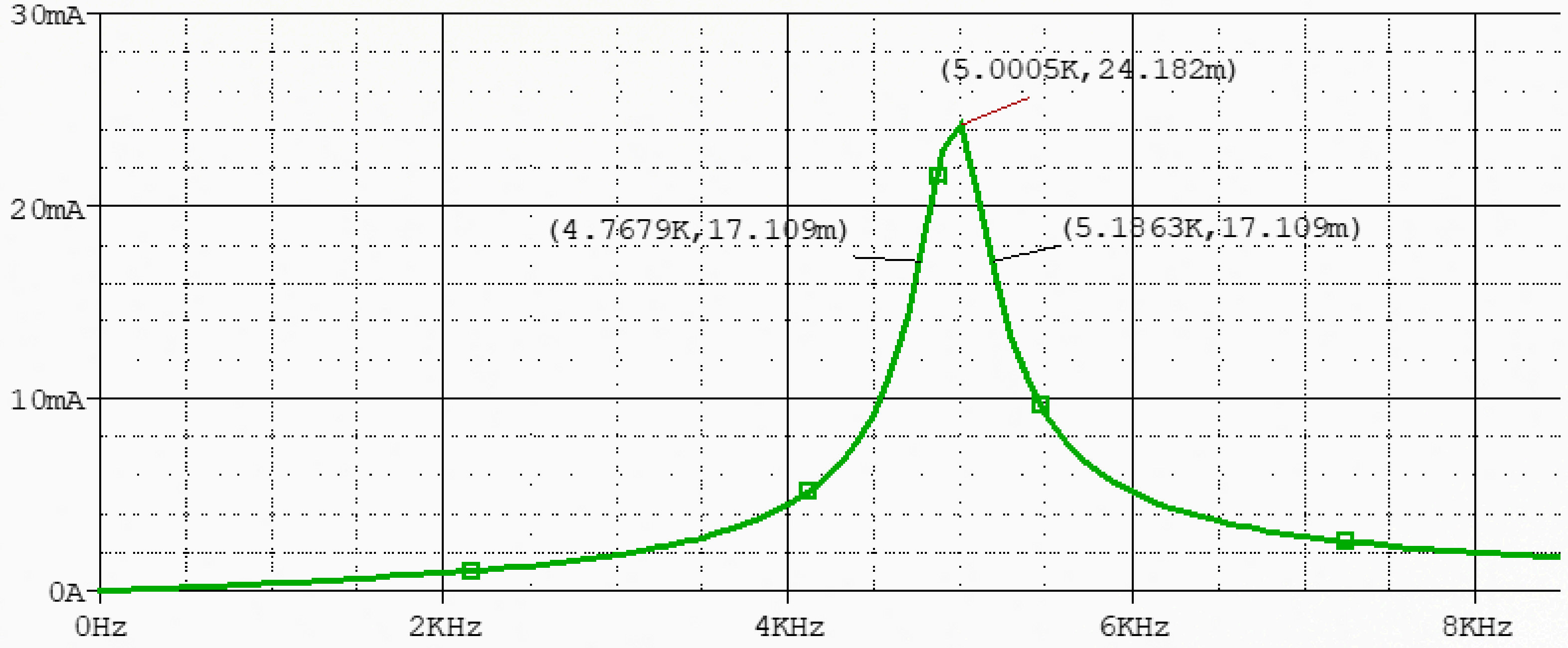
Operation of the RLC Band-pass Filter

- Low Frequency:
  Capacitor blocks current → small output.
- High Frequency:
  Inductor blocks current → small output.
- At Resonance ($f_o \approx 5.0$ kHz):
  - Reactances of L and C cancel each other.
  - Only resistive impedance remains → current is maximum.
  - Voltage across R2 is maximum.
- Bandwidth: Determined by cutoff points f1 & f2 (–3 dB level).









Thank You

Reference:

Boylestad, R. L. (2009). *Introductory circuit analysis* (12th ed.). Pearson Education Ltd.